FINITENESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1874-1884, following the statement at 1864-1872.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Affineness appears at a finite stage. lemma-limit-is-affine (same chapter, card L26293, cited at ~1877-1878). V_i x_{Y_i} X_i is affine for some i.
- Scheme-level descent of finiteness. Limits, Lemma limits-lemma-descend-finite-finite-presentation (cross-chapter, cited at ~1880-1881). V_i x_{Y_i} X_i -> V_i finite for i large.
- Local detection of integral morphisms. Morphisms of Spaces, Lemma spaces-morphisms-lemma-integral-local (cross-chapter, cited at ~1882-1884). f_i finite detected on the etale cover.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZO: focused-seeds/geometry/S9-spaces-limits-descend-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZQ: focused-seeds/geometry/S9-spaces-limits-descend-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If*

(1) *f is finite,*

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *f_0 is locally of finite type,*

then f_i is finite for some $i \geq 0$.

Proof. Choose an affine scheme V_0 and a surjective étale morphism $V_0 \to Y_0$. Set $V_i = V_0 \times_{Y_0} Y_i$ and $V = V_0 \times_{Y_0} Y$. Since f is finite we see that $V \times_Y X = \lim V_i \times_{Y_i} X_i$ is a scheme finite over V. By Lemma 07SQ we see that $V_i \times_{Y_i} X_i$ is affine for some $i \geq 0$. Increasing i if necessary we find that $V_i \times_{Y_i} X_i \to V_i$ is finite by Limits, Lemma 01ZO. For this i the morphism f_i is finite (Morphisms of Spaces, Lemma 03ZQ). □